

Technical Appendix: The Invisible Skeleton: How a 2,000-Year-Old Network Still Decides Who Gets Rich

Introduction and Methodological Rationale

This appendix details the quantitative framework used to test the "Roman Divide" hypothesis: the extent to which old Roman infrastructure predicts 21st-century economic outcomes in England.

The analysis followed a six-stage iterated progression:

1. **Data Curation and Feature Engineering**
2. **Baseline (OLS):** To establish the global relationship between X and Y.
3. **OLS with control variables**
4. **Spatial Error Model (SEM)**
5. **Local Spatial Model (GWR)**
6. **Model Validation**

Data Curation and Feature Engineering

The unit of analysis is the Middle Layer Super Output Area (MSOA, $n=7,201$). Because I had Index of Deprivation data only for England I filtered only the English MSOAs, $n=6856$

Variables

Dependent Variable (Y): 2025 Income Deprivation Score. The variable exhibited a

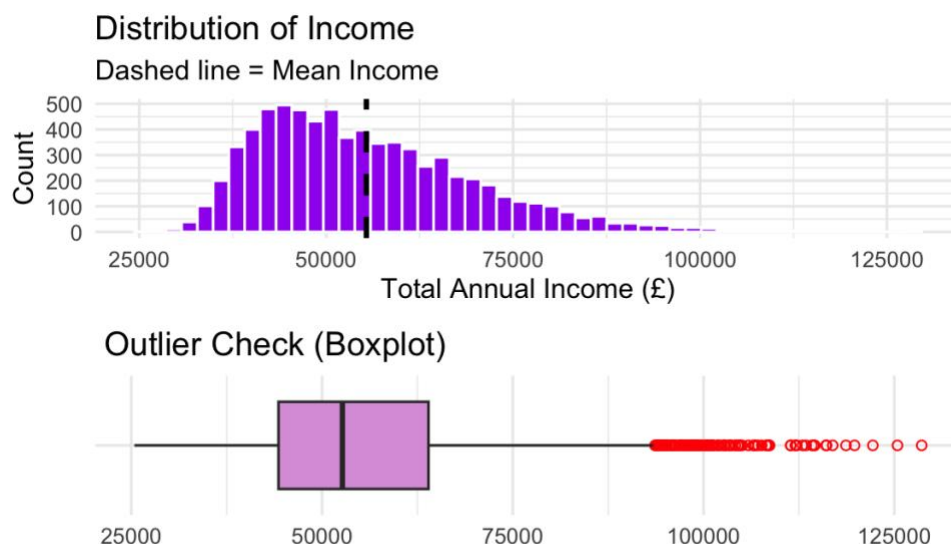


Figure 1 - Histogram and Boxplot for Average Annual Income

severe right-skew and was log-transformed to satisfy normality.

Independent Variable (X): The **Roman Centrality** Score. Defined as follows:

Equation 1 - Roman Centrality Formula

$$RomanCentrality_i = \sum_{j=1}^n e^{-0.09d_{i,j}} * degree(v_j)$$

$$= \sum_{j=1}^n \frac{degree(j)}{e^{0.09d_{i,j}}}$$

where:

n = number of roman junctions = 465 ,

$v = (v_1, v_2, \dots, v_{465})$ = vector of junctions ,

$d_{i,j}$ = distance($MSOA_i$ centroid, v_j),

$degree(v_j)$ = number of roads that create an intersection in junction v_j

To capture the influence of nearby hubs, we applied an exponential distance-decay function. The centrality score divides a junction's degree by $e^{0.09d}$, ensuring that influence diminishes rapidly as distance increases. The decay coefficient 0.09 was calibrated to the [realities of Roman travel](#): a 35km journey (e.g., the 1.2-day march from Londinium to Verulamium) retains meaningful weight, while distant junctions fade to near-zero.

$$\frac{\text{degree}(j)}{e^{\frac{0.09d_{ij}}{4}}} = \frac{4}{e^{0.09*35}} = 0.171$$

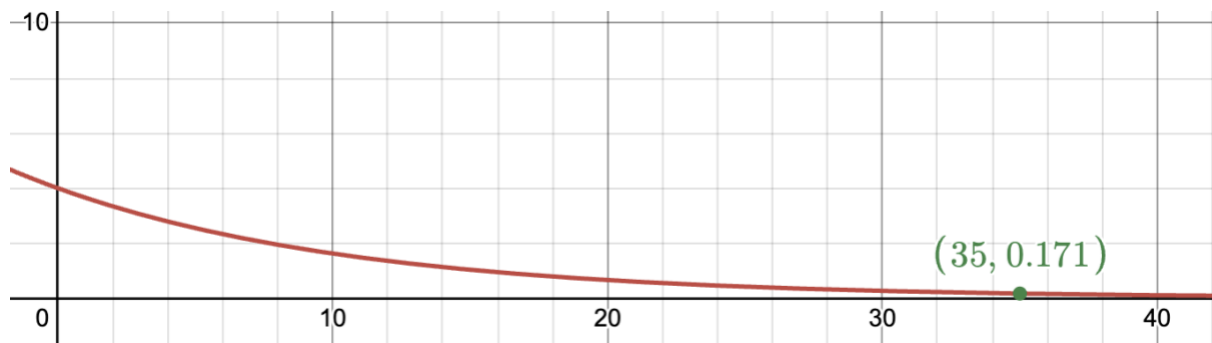


Figure 2 - The Centrality Score formula in $[0,40]$

We can see that this function gets very small quickly, and this is what we want.

Control Variables:

- Education Deprivation Index (source: [English indices of deprivation 2025](#))
- Health Deprivation Index (source: [English indices of deprivation 2025](#))
- Min Distance from centroid to 2025 Motorway (geographic analysis based on [Major Road Network database](#))
- Percentage of population (16+) that work long hours (49 hours or more in a week) (based on [Census 2021](#))
- Population Density (based on [English indices of deprivation 2025](#))
- Distance to London (geographic analysis)

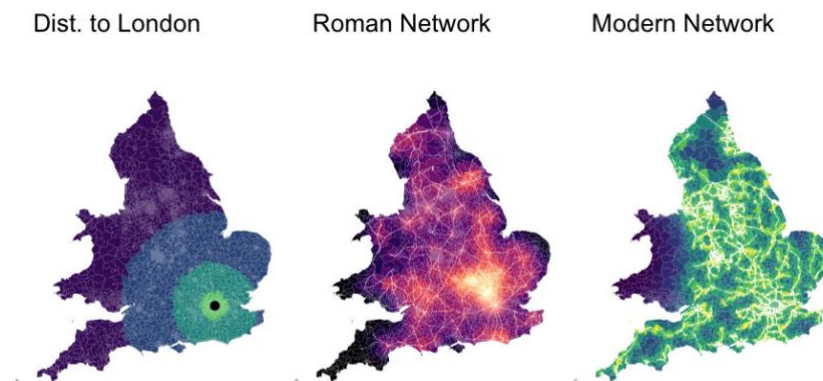


Figure 3 - distance to: London, Roman Roads, Modern Road Network

The IoD25 index was on the LSOA level, so I applied weighted average on the population to “group by” MSOA.

Some variables needed to be “logged” to address skewness, For example - this Roman Centrality after the log:

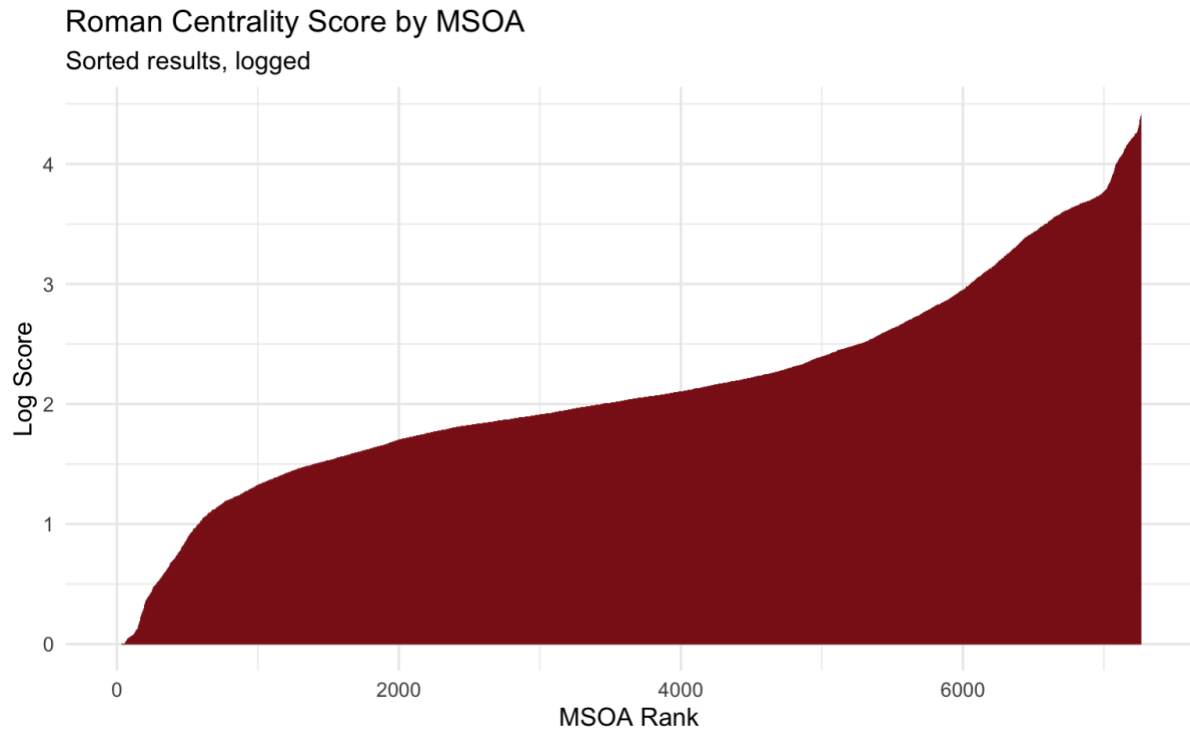


Figure 4 - Roman Centrality by MSOA, log scale

Phase 0: Baseline Model

We first estimated the standard Ordinary Least Squares specification:

Equation 2 - Baseline Model

$$\log(\text{AverageIncome}) = \log(\text{RomanCentrality})$$

```
Call:
lm(formula = log_income ~ log_adj_decay_score, data = msoas_model_ready)

Residuals:
    Min       1Q   Median       3Q      Max
-0.69395 -0.15434 -0.00064  0.14404  0.73505

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    10.568534   0.006553  1612.74  <2e-16 ***
log_adj_decay_score  0.150975   0.002847   53.04  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2108 on 7262 degrees of freedom
Multiple R-squared:  0.2792,    Adjusted R-squared:  0.2791
F-statistic: 2813 on 1 and 7262 DF,  p-value: < 2.2e-16
```

Figure 5 - Baseline Model R Results

While the R^2 was relatively high, we should not conclude anything from this model, because in our case, clearly, our dependent variable is much more likely to be explained by other, more reasonable variables (such as education or health). This was done only to get something to compare to.

Phase 1: OLS with control variables

Equation 3 - OLS with control variables

$$\log(\text{AvgIncome}) = \beta_0 + \beta_1 \log(\text{RomanCentrality}) + \beta_2 \log(\text{PopDensity}) + \beta_3 \text{EducationDeprived} + \beta_4 \text{HealthDeprived} + \beta_5 \log(\text{DistMotorway}) + \beta_6 \text{PctLongHours}$$

This time I added all the control variables. I did expect a much higher R^2 , but I was more interested in what will happen to the roman **coefficient**.

```
lm(formula = log_income ~ log_adj_decay_score + log(pop_density) +
  education_score + health_score + log_dist_motorway + log_dist_london +
  pct_long_hours, data = final_data_boss)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.52365	-0.05955	-0.00004	0.05786	0.51584

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	11.3100041	0.0215877	523.909	< 2e-16	***
log_adj_decay_score	0.0082241	0.0018129	4.537	5.82e-06	***
log(pop_density)	0.0099339	0.0011446	8.679	< 2e-16	***
education_score	-0.0046411	0.0001206	-38.475	< 2e-16	***
health_score	-0.1334152	0.0026798	-49.785	< 2e-16	***
log_dist_motorway	-0.0162892	0.0017701	-9.203	< 2e-16	***
log_dist_london	-0.0908065	0.0016987	-53.457	< 2e-16	***
pct_long_hours	0.0018467	0.0004801	3.846	0.000121	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09179 on 6848 degrees of freedom

Multiple R-squared: 0.8642, Adjusted R-squared: 0.864

F-statistic: 6224 on 7 and 6848 DF, p-value: < 2.2e-16

Figure 6 - OLS with control variables model - R Results

Our variable's (log_adj_decay_score) coefficient decreased dramatically. The signs of the coefficients behaved as I expected. According to this model 1% increase in roman centrality is associated with a 0.008% **increase** in income, which I found very low. While the coefficient for Roman Gravity was statistically significant, the validity of OLS relies on the assumption that errors are independent.

However:

Assumption 5 Check: Are Errors Clustered?

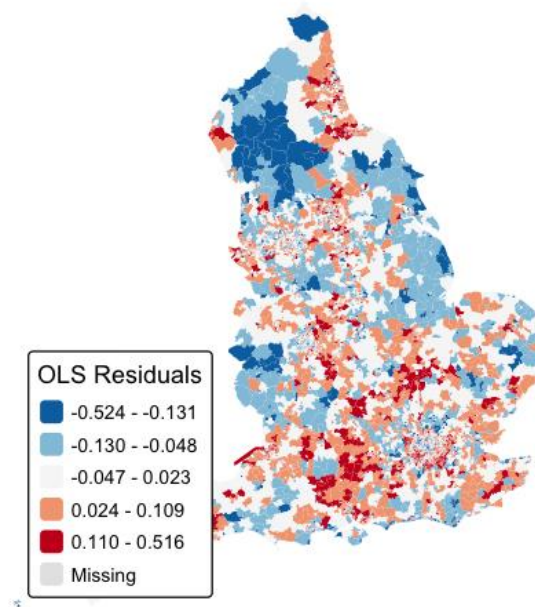


Figure 7 - Mapping the OLS Residuals

```
[1] "--- ASSUMPTION 5: SPATIAL INDEPENDENCE (Moran's I) ---"

Global Moran I for regression residuals

data:
model: lm(formula = log_income ~ log_adj_decay_score + log(pop_density) + education_score +
health_score + log_dist_motorway + log_dist_london + pct_long_hours, data = gwr_data_sf)
weights: lw_clean

Moran I statistic standard deviate = 52.684, p-value < 2.2e-16
alternative hypothesis: two.sided
sample estimates:
Observed Moran I      Expectation      Variance
3.844265e-01      -8.577078e-04      5.348155e-05
```

Figure 8 - Moran's I R results

Moran's I Test on Residuals: $I=0.38$, statistically significant

This confirms the presence of Spatial Autocorrelation: the model's errors are not random but are spatially clustered.

Following Anselin (2008), **Lagrange Multiplier** tests identified the Spatial Error Model (SEM) as the superior specification. The significant Robust LM-Error statistic confirms that spatial dependence is driven by omitted structural factors (like unobserved history) rather than direct contagion between neighbors, which also seemed reasonable in our scenario.

Phase 2: The Spatial Error Model (SEM)

To address the violation of independence detected in Phase 1, we estimated a Spatial Error Model. Unlike OLS, the SEM assumes that the error terms themselves are spatially correlated, likely due to spatially clustered omitted variables.

Results & Interpretation:

```
Call:errorsarlm(formula = final_formula, data = final_data_boss, listw = lw_clean,
  method = "LU", zero.policy = TRUE)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.53465357	-0.04874408	-0.00018167	0.04779810	0.49783103

Type: error

Regions with no neighbours included:

6531

Coefficients: (asymptotic standard errors)

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	11.17870395	0.03015591	370.6969	< 2.2e-16
log_adj_decay_score	0.02032889	0.00401086	5.0685	4.010e-07
log(pop_density)	0.00720148	0.00123631	5.8250	5.713e-09
education_score	-0.00359620	0.00013387	-26.8639	< 2.2e-16
health_score	-0.13457827	0.00332639	-40.4578	< 2.2e-16
log_dist_motorway	-0.01339211	0.00218062	-6.1414	8.179e-10
log_dist_london	-0.08213181	0.00324720	-25.2932	< 2.2e-16
pct_long_hours	0.00750065	0.00058052	12.9206	< 2.2e-16

Lambda: 0.65468, LR test value: 2117.7, p-value: < 2.22e-16

Approximate (numerical Hessian) standard error: 0.011374

z-value: 57.559, p-value: < 2.22e-16

Wald statistic: 3313, p-value: < 2.22e-16

Log likelihood: 7708.651 for error model

ML residual variance (sigma squared): 0.0056192, (sigma: 0.074961)

Number of observations: 6856

Number of parameters estimated: 10

AIC: -15397, (AIC for lm: -13282)

Figure 9 - SEM R Results

The Spatial Error Model vastly outperformed the OLS baseline (AIC improved from -13,282 to -15,397), confirming strong spatial spillovers ($\lambda=0.65$, $p<0.001$). However, the global coefficient remained weak ($\beta=0.02$). This suggests that a national average dilutes the signal and motivates the shift to a local model (GWR) to unmask region-specific impacts.

Phase 3: Geographically Weighted Regression (GWR)

Results

Explanatory Power: The model demonstrates a strong fit, with a median local R^2 of 0.85, replacing the single global metric.

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4808	0.8137	0.8534	0.8437	0.8863	0.9479

Figure 10 - Statistics of local R -squared

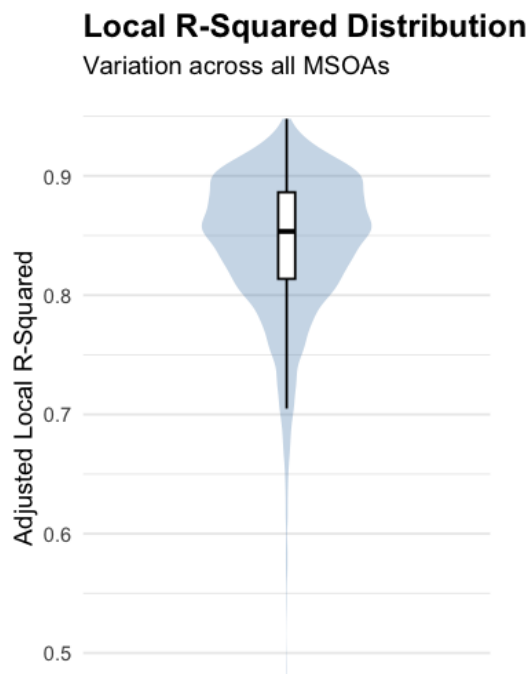


Figure 11 - Local R -squared distribution

Roman Centrality Coefficient

Extreme outliers in Cornwall were identified as statistical noise caused by the region's lack of Roman infrastructure:

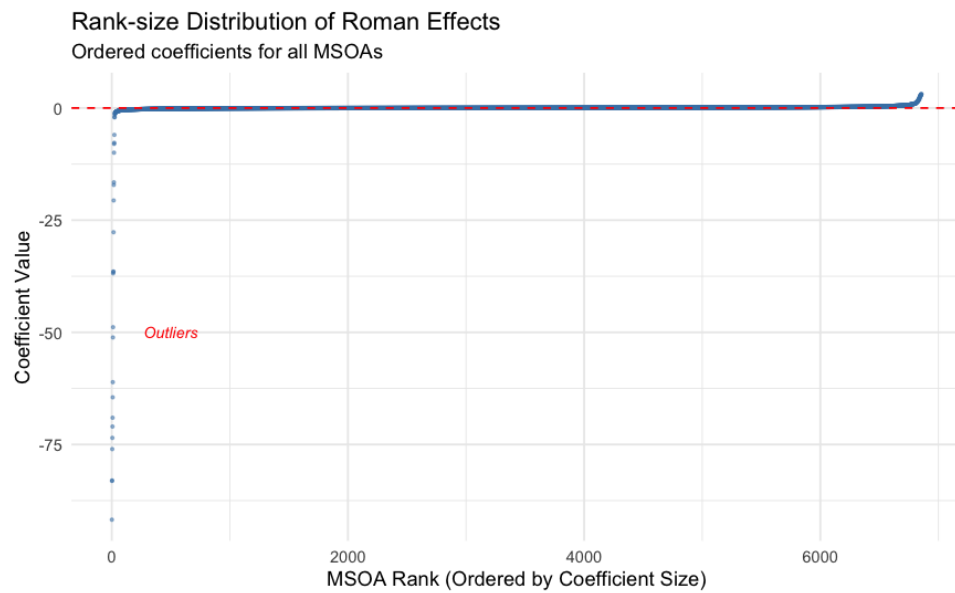


Figure 12 - Roman Centrality Beta Coefficients, ranked by size

To ensure the map reflected only valid relationships, I applied a dual-filtering strategy: I masked results that failed the 95% significance threshold and excluded areas with negligible Roman centrality. This effectively removed artifacts from unconnected regions, isolating the genuine patterns revealed by the model:

Where History Matters: Local Roman Coefficients

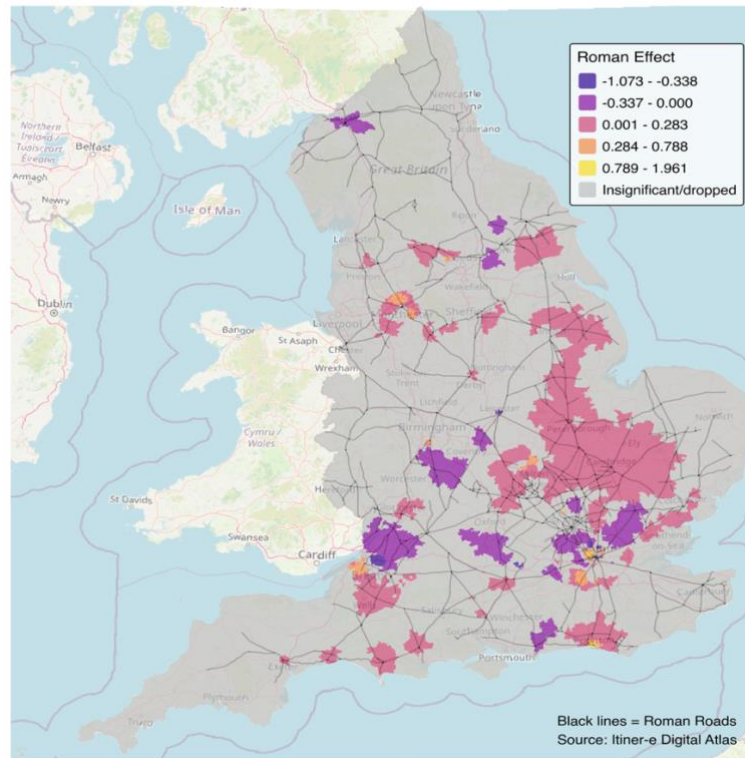


Figure 13 - National GWR beta coefficients

Our final AIC results were:

Local GWR AICc: -15843.6

Which is another improvement of 446.3 points.

Final Model Selection

The GWR provided the best fit of all three models. The massive reduction in AICc (compared to both OLS and SEM) confirms that the relationship between ancient roads and modern wealth is structurally non-stationary. It is not just that errors cluster, but that the mechanism itself changes across the map.

Validating the final model assumptions:

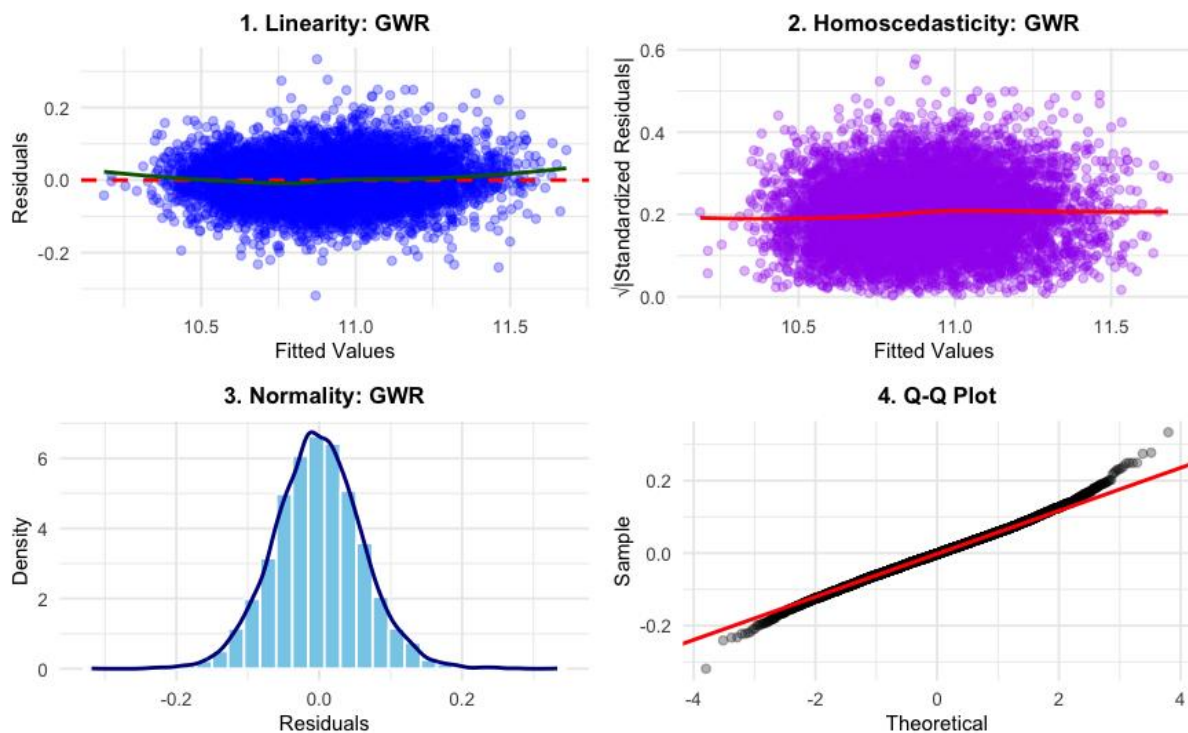


Figure 14 - Linear Regression Model validations

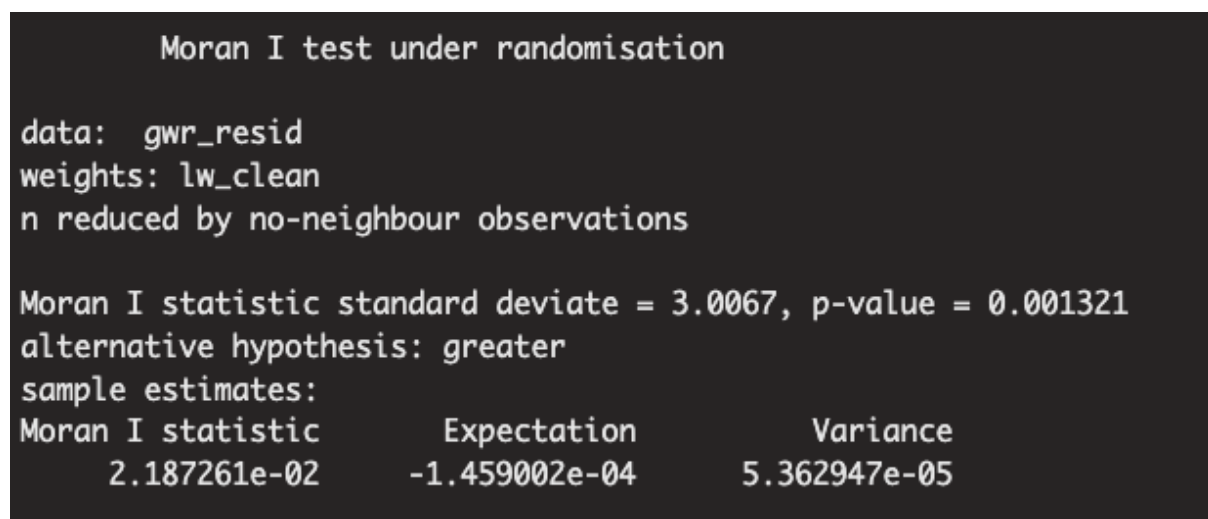


Figure 15 - Moran's I for GWR - R output

Moran's I is close to zero - as we wanted.

Summary and Future Work

Our results reveal a double-edged legacy. While the global model confirms a baseline association, the GWR unmasks a distinct split: ancient infrastructure acts as a "wealth multiplier" along the Ermine Street spine and commuter belts, but inverts into a "path-dependency trap" in industrial hubs like Leicester.

Two key constraints remain:

- Unobserved Mediators: The model lacks intermediate historical variables (e.g., the Industrial Revolution), leaving the specific causal mechanism a "black box."
- Aggressive Filtering: Our dual-filtering strategy, while effective at removing noise in unconnected regions, was likely too strict, potentially discarding valid signals in areas of intermediate Roman influence.

Future Work Future research should consider pivoting from GWR to functional clustering. I propose using K-Means to classify MSOAs into typologies (e.g., Urban vs. Rural) and applying a Mixed Effects Model.

Acknowledgement:

I acknowledge the use of Google Gemini (Model: Gemini 3) to help structure the arguments and to generate initial code snippets in this project. The specific prompts used involved asking for the syntax of certain R methods, library names and asking for a critique of my methodology. The output from the AI was reviewed, fact-checked, and modified by me to ensure accuracy and originality.

Student Number:

25190920